

eVersum

SUMS Handbook

Software Update Management System according to UN Regulation No. 156

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Handbook ID / chapter numbering: [TO BE CONFIRMED] — to be aligned with the existing SUMS documentation (SUMS_Dokumentenmanagement-Prozess)

Scope limitation — Over-the-air (OTA) updates are excluded. eVersum vehicles receive software updates exclusively in the workshop, performed by eVersum technicians. Paragraph 7.1.4 of UN R156 is not applicable. This statement applies to this handbook and to all reports generated by the SUMS tool.

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1 Document control

Version	Date	Author	Change
0.1 (draft)	25 September 2026	R. Adler	First draft — describes the SUMS tool and maps it to UN R156 §7.1 and the TÜV SÜD self-assessment.
0.2 (draft)	25 September 2026	R. Adler	Exports for authorities (register PDF, CSV), CSV import for migration (5.9), security controls extended (5.8, 4.9).

Role	Name	Signature / date
Prepared	R. Adler	
Reviewed	J. Zdun, Head of Systems and Security [TO BE CONFIRMED]	
Approved	[TO BE CONFIRMED]	

2 Introduction

2.1 Purpose

This handbook describes how eVersum manages software updates of its vehicles in accordance with UN Regulation No. 156 (Software Update Management System, SUMS). It explains which processes are used, who is responsible for them, which records are kept and how the eVersum SUMS tool supports and enforces those processes. It is intended as the reference document for the Approval Authority and the Technical Service (TÜV SÜD) and as the working instruction for the eVersum staff involved.

The handbook follows the structure of the TÜV SÜD self-assessment for UN R156. Chapter 7 maps every requirement of §7.1 to the process step and to the evidence produced by the tool.

2.2 Scope

- Vehicle types: e-Shuttle MK II-400 (model code ES03) **[TO BE CONFIRMED]** — list all vehicle types in scope of the type approval.
- Software in scope: all software relevant to a type-approved system and grouped under an RXSWIN, for all ECUs listed in the ECU register (in-house and supplier ECUs).
- Update method: re-flashing in the workshop by eVersum technicians. No over-the-air updates; no updates performed by the vehicle user.
- Out of scope: diagnostic functions, OTA (§7.1.4), vehicle production processes other than recording the end-of-line configuration.

2.3 Replacement of the previous implementation

The SUMS was previously implemented in Helix ALM (RXSWIN documents, Initial End of Line Configuration and Software Update documents with baselining). The eVersum SUMS tool implements the same data model — see the R156 SUMS Overview diagram — so that every Helix record type has a one-to-one counterpart. The ERP system (work orders, collections of vehicles) and the Egnyte software repository remain in use and are linked from the tool.

Migration of existing Helix baselines: **[TO BE CONFIRMED]** — either an export of the current live baselines or a defined cut-off date from which records are kept only in the SUMS tool.

2.4 References

Reference	Document
[R156]	UN Regulation No. 156 — Uniform provisions concerning the approval of vehicles with regards to software update and software update management system
[ID]	Interpretation Document for UN R155/R156 (ECE/TRANS/WP.29/2025/45)
[ISO 24089]	ISO 24089:2023 Road vehicles — Software update engineering
[ISO 27001]	ISO/IEC 27001 Information security management systems
[POL]	M1-DO-007 Cybersecurity Management Policy (eVersum, 15.04.2025)
[DMP]	SUMS Dokumentenmanagement-Prozess (eVersum, 01.07.2025)
[SA]	TÜV SÜD UNR156 SUMS Self-assessment eVersum v1.0
[OVW]	R156 SUMS Overview (eVersum diagram)

2.5 Terms and abbreviations

Term	Meaning
RXSWIN	R Series Software Identification Number — identifier of the software relevant to a type-approved system of a vehicle type (§2.3 R156)
Baseline	Numbered, released state of all software belonging to one RXSWIN (versions, files, SHA-256 checksums, compatible hardware, change log)
SU document	Software Update document — record of one software update according to §7.1.2.5
EOL configuration	Initial End of Line Configuration — software and hardware configuration of a vehicle when it leaves production

Term	Meaning
LKC	Last Known Configuration — the most recent recorded configuration of a vehicle (VIN)
SHA-256	Cryptographic hash used as integrity validation data of software and configuration files
V&V	Verification and validation
ECU	Electronic control unit
ERP WO	Work order in the eVersum ERP system

3 SUMS overview

3.1 Systems

System	Content	Role in the SUMS
eVersum SUMS tool	Vehicle types, ECU register, RXSWIN register with baselines, Software Update documents, target vehicles, vehicle configurations (EOL / LKC), audit trail	System of record for all R156 information; enforces the process rules
Egnyte software repository	Software and configuration files per ECU, readme files	Secure storage of the binaries; referenced by hyperlink from the SUMS tool
ERP system	Collections of vehicles, work orders per VIN	Production and service orders; linked by work order number; reads the LKC from the SUMS tool

Hosting of the SUMS tool: **[TO BE CONFIRMED]** — currently a preview instance on a dedicated server (EU, Hetzner); production operation on the eVersum local server. Coverage of the SUMS tool server by the eVersum ISO 27001 ISMS **[TO BE CONFIRMED]**.

3.2 Roles and responsibilities

Role in the tool	Typical function	Permissions
Administrator	Head of Systems and Security [TO BE CONFIRMED]	All functions; user management; release of baselines and SU documents; V&V sign-off
QC manager	Engineering lead / QC [TO BE CONFIRMED]	Release of baselines and SU documents; V&V sign-off; all editing functions
Technician	Software / service engineer, workshop technician	Create and edit drafts (baselines, SU documents); SHA-256 verification before flashing; recording of update results; recording of EOL configuration
Partner viewer	External party with read access	Read only

Every user has a personal account; shared accounts are not permitted. Every change in the tool is recorded in the audit trail with the user, time, and the state before and after (see 4.9). User accounts are created by an administrator with a random initial password.

Authentication: local accounts **[TO BE CONFIRMED]** — integration with eVersum AD / SSO to be decided.

3.3 Process overview

1. **Register** the vehicle type, its ECUs and the RXSWINs (chapter 5.1).
2. **Release a software baseline** for an RXSWIN when new software is ready: versions, files, SHA-256, compatible hardware, change log; publish the readme to Egnyte (5.2).
3. **Document the software update** in an SU document: purpose, dependencies, affected RXSWINs, type approval assessment, execution and safety conditions, V&V sign-off, target vehicles with compatibility confirmation; then release it (5.3).
4. **Execute** the update in the workshop: verify the file SHA-256 against the register, flash, record the result per vehicle — the tool records the new Last Known Configuration (5.4).
5. **Notify** the vehicle user / fleet manager and record the notification (5.5).
6. **Provide information** to authorities on request from the tool (5.7).

4 Records kept in the SUMS tool

All records below are held in a single PostgreSQL database of the SUMS tool. Records that represent a released state are protected against modification at database level (database triggers), not only by the user interface; a change always creates a new record or a new revision, and the previous one remains readable.

4.1 Vehicle type

Name and model code of the vehicle type. RXSWINs, ECUs and SU documents are defined per vehicle type.

4.2 ECU register (§7.1.1.2)

Per vehicle type: ECU name, system, supplier, eVersum part number and the UN-ECE regulation the ECU is relevant to.

4.3 Installed ECUs per vehicle

Per VIN and ECU: serial number, hardware version and batch number. A change after the end-of-line configuration automatically creates a new Last Known Configuration.

4.4 RXSWIN register (§7.1.2.3)

Each RXSWIN is a record with its identifier, the vehicle type, a description of the system functionality and the regulations affected. Its software is described by numbered baselines:

Field (per ECU in a baseline)	Content
ECU / part number / supplier	From the ECU register
SW version, SW file name	Unique identification of the software
SW file SHA-256	Integrity validation data (method: SHA-256)
Config version, config file, config SHA-256	Configuration / parameter file, if applicable
Compatible hardware	Hardware precondition (§7.1.1.7)
Egnyte folder	Hyperlink to the files in the software repository
Change log, description	Content of the change

Baseline life cycle: **draft** (editable) → **released** (read-only) → **superseded** (when the next baseline is released; remains visible and exportable). A baseline can only be released by an administrator or QC manager and only if every file has a valid SHA-256 checksum. A change to released software always requires a new baseline, which starts as a copy of the current released one.

4.5 Software Update document (§7.1.2.5)

Field	R156 reference
Document ID (SU-<year>-<no.>), revision, title	Unique identification
Purpose of the update	§7.1.2.5 (a)
Dependencies identified, system schemes baseline, new hardware required	§7.1.1.5, §7.1.2.5 (b), (g)
Affected RXSWINs with baseline before and after	§7.1.2.3, §7.1.2.5 (c)
Type approval update necessary (yes/no) + justification	§7.1.1.8–10, §7.1.2.5 (d), (e)
Affected UN-ECE requirements / tests	§7.1.1.8, §7.1.2.5 (d), (e)
Approval granted, approval number and date	§7.1.2.5 (f)
Execution conditions, safe state conditions	§7.1.2.5 (g)
Actions required from the user / a competent person	§7.1.2.5 (g)
Safety / security confirmation	§7.1.2.5 (h)

Field	R156 reference
V&V status (pass/fail), method and evidence, engineering sign-off (name, time)	§7.1.3.3, §7.1.2.5 (i)
User notification required; notified by / when / how	§7.1.1.11
Target vehicles with compatibility confirmation and execution result	§7.1.1.6, §7.1.1.7, §7.1.2.4
ERP work order (number and link), Egnyte folder	Links to ERP and repository

SU document life cycle: **draft** → **released** → **superseded** (by a new revision). The tool refuses the release until all required fields are complete, the V&V is passed and signed, every affected RXSWIN points to a released baseline and every target vehicle has a confirmed compatibility. After release only two things can be recorded: the execution result per target vehicle and the user notification.

4.6 Target vehicles (§7.1.2.4)

List of VINs per SU document. Only vehicles of the SU document's vehicle type can be added. For each vehicle the tool shows its Last Known Configuration and whether the installed baseline matches the expected 'baseline before' of every affected RXSWIN (match / mismatch / already installed / no configuration recorded). Compatibility is confirmed per vehicle with name and time.

4.7 Vehicle configurations (§7.1.2.2)

Immutable snapshots per VIN: the **Initial End of Line configuration** (once per vehicle) and every subsequent **Last Known Configuration**. A snapshot contains the installed baseline of every RXSWIN with all software versions and SHA-256 checksums and the installed ECUs with serial number, hardware version and batch. A new LKC is recorded automatically after each successful software update (reason: SU document and revision) and after each change of installed ECU hardware.

Type approval relevant vehicle or system parameters (§7.1.2.2, 4th criterion) **[TO BE CONFIRMED]** — define which parameters are to be recorded with the configuration.

4.8 Reports

- **Readme** per ECU software in a baseline — same structure as the Helix readme; stored in the ECU folder on Egnyte.
- **Software Update report** (PDF) — the complete SU record per §7.1.2.5 including target vehicles and results.
- **Last Known Configuration** per VIN — in the tool and through the ERP interface (JSON).
- **RXSWIN register** (PDF) — all RXSWINs of a vehicle type with every baseline (including superseded ones), all software versions and SHA-256 checksums, and the Software Update documents that changed them.
- **Vehicle configurations** (CSV) — the Last Known Configuration of every vehicle, one row per VIN and ECU.
- **Audit trail** — filterable by record, action, user and date; exportable as CSV.

4.9 Audit trail

Every creation, change, release, supersession, signature, verification, execution, notification, import, failed login and ERP read access is written to the audit trail with: user (name), action, record, time, state before and after, and client IP address.

The audit trail is append-only and this is enforced by the database: the application connects with a restricted database role that may only read and insert audit entries, and a database trigger rejects any update or deletion. The restricted role is not the owner of the tables and therefore cannot disable the protection triggers of released records either.

5 Processes

5.1 Registering vehicle types, ECUs and RXSWINs

Responsible	Administrator / QC manager (vehicle types); technician or engineer (ECUs, RXSWINs)
Input	Type approval documentation, list of ECUs relevant to type-approved systems
Output	Vehicle type, ECU register, RXSWINs with description and regulations affected

1. Create the vehicle type (name, model code).
2. Register every ECU relevant to a type-approved system: name, system, supplier, eVersum part number, UN-ECE regulation.
3. Create one RXSWIN per **[TO BE CONFIRMED]** (per regulation or per system), with description of the functionality and the regulations affected. RXSWIN numbering convention: **[TO BE CONFIRMED]** (example: R48SWIN001).

5.2 Releasing a software baseline

Trigger	New or changed software / configuration file for an ECU belonging to an RXSWIN
Responsible	Engineer (draft); administrator or QC manager (release)
Output	Released baseline; readme PDF on Egnyte

1. Open a new baseline for the RXSWIN. It starts as a copy of the current released baseline.
2. For each changed ECU enter the SW version, file name, configuration version, compatible hardware, Egnyte link and change log.
3. Compute the SHA-256 of the software (and configuration) file with the built-in tool. The file is hashed locally in the browser and is not uploaded; the result is stored in the baseline.
4. Peer review of the draft baseline (change log, compatible hardware, checksums).
5. Release the baseline. The tool checks that every file has a valid SHA-256. The baseline becomes read-only; the previous one becomes superseded.
6. Download the readme PDF of each changed ECU and store it in the ECU software folder on Egnyte next to the software file.

5.3 Software Update document

Trigger	Decision to update vehicles in the field or in production to a new baseline
Responsible	Engineer (draft); administrator / QC manager (V&V sign-off, release)
Output	Released SU document; Software Update report (PDF)

1. Create the SU document for the vehicle type: title and purpose of the update.
2. Identify dependencies: systems and interfaces affected, based on the latest system schemes baseline; state whether new hardware is required.
3. Add the affected RXSWINs and the new baseline for each; the tool records the baseline before.
4. Assess the effect on type approval: decide whether a type approval update is necessary and justify the decision; list the affected UN-ECE requirements / tests; if an update of the approval is necessary, record the approval number and date once granted.
5. Define the execution: conditions, safe state, actions required from the user or a competent person, and the confirmation that the update is conducted safely and securely.
6. Add the target vehicles (VINs). Check the Last Known Configuration shown for each vehicle against the expected baseline before and against the compatible hardware in the baseline; confirm compatibility per vehicle or remove the vehicle.
7. Perform verification and validation; the administrator / QC manager records pass or fail with method and evidence (engineering sign-off).
8. Release the SU document. The tool refuses the release if any condition is missing and lists what is missing.

9. A later change requires a new revision; V&V and compatibility confirmations are repeated.

5.4 Execution of the update in the workshop

Precondition	Released SU document with the vehicle listed as target
Responsible	eVersum technician
Output	Recorded result; new Last Known Configuration

1. Establish the execution conditions and the safe state defined in the SU document.
2. Obtain the software file from the Egnyte repository only.
3. Before flashing, use 'Verify file' on the baseline item: the tool computes the SHA-256 of the file and compares it with the register. Only a MATCH result allows flashing. Every verification (match or mismatch) is recorded in the audit trail with the technician's name.
4. Flash the ECU. Verify the software version on the component **[TO BE CONFIRMED]** (CAN tool / read-out procedure, §7.1.1.4).
5. Record the result (success / failed / rolled back) for the VIN in the SU document. A success automatically records the new Last Known Configuration of the vehicle. A recorded result cannot be overwritten.

5.5 User notification

If the SU document requires notification, the fleet manager / customer is informed by e-mail about the update and its purpose — one notification per SU document. The person who sent the notification records it in the SU document (method, time). The tool records who made the entry.

Whether the tool sends the e-mail itself or only records it: **[TO BE CONFIRMED]**.

5.6 End-of-line configuration

When a vehicle leaves production, the technician records the Initial End of Line configuration: the installed baseline of every RXSWIN, the configuration ID, the system schemes baseline, the V&V status and the ERP work order. The installed ECUs (serial number, hardware version, batch) are entered per vehicle. The EOL configuration can be recorded only once and is read-only.

5.7 Information for the Approval Authority and the Technical Service

On request, the following information is provided directly from the tool: the RXSWIN register with all baselines (RXSWIN register PDF, readme PDFs), the SU documents with target vehicles and results (Software Update report PDF), the Last Known Configuration of every vehicle (CSV) or of a single VIN, and the audit trail (CSV). Access for an assessor can be granted through a read-only account.

5.8 Access management, integrity and backup

- Personal user accounts with role-based permissions (3.2); accounts are deactivated, not deleted, when a person leaves. Passwords of at least 12 characters; users change their own password, administrators can reset it to a one-time random password.
- Deactivation, role changes and password changes take effect immediately: every request is checked against the user record; access tokens are valid for 15 minutes; the long-lived session token is held in an HttpOnly cookie that scripts cannot read.
- Login protection: limited number of failed login attempts; failed logins are recorded in the audit trail; HTTPS only; server firewall allows only HTTPS and SSH (key-based).
- Released records and the audit trail are protected against modification by database triggers; the application uses a restricted database role that cannot disable them.
- Links to external systems (Egnyte, ERP) accept only http(s) addresses; CSV exports are protected against spreadsheet formula injection.
- Database backup: automatic daily dump at 03:30, verified before it replaces the previous copy, kept for 14 days on the server. Off-site copy, retention period for production and periodic restore test **[TO BE CONFIRMED]**.
- ERP read access uses a dedicated API key stored only on the server.

5.9 Migration and bulk import

Existing records can be imported from CSV files (e.g. an ERP export of VINs or a Helix export of baseline data). Every import shows a preview first; nothing is saved until the user confirms it, and the import is only possible when every row is valid (all-or-nothing). Each import is recorded in the audit trail with the imported content.

- **Vehicles:** VIN, name and year per vehicle type. VINs already registered are skipped.
- **Baseline items:** one row per ECU (name or eVersum part number) with software version, files, SHA-256 checksums, compatible hardware, Egnyte link and change log — only into a draft baseline, which is then reviewed and released as described in 5.2.

6 Compliance mapping to UN R156 §7.1

The table maps each requirement to the process and the evidence produced. The column 'Open' lists what eVersum still has to confirm or provide; these items are also collected in chapter 7.

Requirement	How it is fulfilled	Evidence	Open
7.1.1.1 Secure storing and sharing	All R156 information is held in the SUMS tool (single database) with role-based access, personal accounts, HTTPS, audit trail and database-level protection of released records. Software binaries on Egnite with restricted permissions. Information can be provided on request (5.7).	Chapter 4, 5.7, 5.8; audit trail; access roles	ISO 27001 scope incl. tool server [TO BE CONFIRMED]; backup policy [TO BE CONFIRMED]
7.1.1.2 Identification of software versions	ECU register + baselines: SW version, file name, SHA-256 of SW and config file, compatible hardware; per vehicle: ECU serial number, HW version, batch.	RXSWIN register, readme PDF, vehicle configuration	
7.1.1.3 Management of RXSWIN information	RXSWIN held at the manufacturer. Before/after state: SU document records baseline before and after per affected RXSWIN; LKC per VIN shows the installed baseline. Update of information = new baseline (5.2).	SU document, baselines, LKC	RXSWIN readable from the vehicle (OBD)? [TO BE CONFIRMED] — if not, §7.2.1.2.2 requires declaration of software versions to the Approval Authority
7.1.1.4 Verification of software versions	Expected versions and checksums per VIN are available from the LKC. File verification by SHA-256 before flashing is recorded. On-vehicle read-out with CAN tools.	LKC, verification entries in audit trail	Read-out procedure / tool [TO BE CONFIRMED]
7.1.1.5 Interdependencies	SU document: dependencies identified, system schemes baseline, affected RXSWINs; V&V sign-off.	SU report sections 1–3	Source and ID format of the system schemes baseline [TO BE CONFIRMED]
7.1.1.6 Identification of target vehicles	Target vehicles on VIN level per SU document. Error prevention: only vehicles of the correct vehicle type can be selected; the LKC of each vehicle is compared with the expected baseline before; compatibility confirmed per VIN; ERP work order linked.	SU report section 7	
7.1.1.7 Confirmation of compatibility	Before release: per VIN the LKC (software and hardware) is shown against the baseline before; compatible hardware listed in the baseline and readme; confirmation per VIN with name/time; release blocked until all are confirmed; regression testing covered by V&V sign-off.	SU report section 7; audit trail	
7.1.1.8 Effect on type approval	Required decision 'type approval update necessary' with justification; affected UN-ECE requirements / tests; release blocked without them.	SU report section 4	
7.1.1.9 Changed functionality	Same assessment as 7.1.1.8; justification must address information package, test results and functions.	SU report section 4	Guidance text for the justification [TO BE CONFIRMED]
7.1.1.10 Changes compared to registration	Dependencies and type approval assessment in the SU document.	SU report sections 1 and 4	
7.1.1.11 User information	Notification required flag; notification recorded (who, when, method); one notification per SU	SU report section 6	E-mail sent by the tool or only

Requirement	How it is fulfilled	Evidence	Open
	document; no user action needed (workshop update).		recorded [TO BE CONFIRMED]
7.1.1.12 Information to authorities	RXSWIN register (PDF), SU documents with target vehicles (PDF), vehicle configurations and audit trail (CSV) are exported directly from the tool.	5.7	
7.1.2.1 Documentation of processes	This handbook; processes 5.1–5.8 apply to every vehicle type in scope.	This document	List of vehicle types [TO BE CONFIRMED]
7.1.2.2 Configuration before/after update	EOL configuration and LKC snapshots per VIN with SW versions, SHA-256, ECU hardware identification; SU document records baseline before/after.	Vehicle configuration history	Type approval relevant vehicle / system parameters [TO BE CONFIRMED]
7.1.2.3 RXSWIN register	Auditable register: baselines numbered, released baselines read-only, superseded baselines kept; SW versions and SHA-256 for all software of each RXSWIN; method SHA-256; SU document lists affected RXSWINs.	RXSWIN register, audit trail	
7.1.2.4 Target vehicles	VIN list per SU document with compatibility confirmation and execution result.	SU report section 7	
7.1.2.5 (a)–(f)	Purpose; dependencies; affected RXSWINs (type-approved systems); type approval decision + justification; affected requirements; approval granted with number and date.	SU report sections 1, 2, 4	
7.1.2.5 (g)	Execution conditions; new hardware required; safe state; actions required from the user / a competent person.	SU report sections 1, 5	
7.1.2.5 (h)	Safety / security confirmation with justification of how the conditions are met.	SU report section 5	
7.1.2.5 (i)	V&V status pass/fail, method and evidence incl. adequacy of the method, engineering sign-off with name and time.	SU report section 3	
7.1.3.1 Prevention of manipulation	SHA-256 recorded at release; file verified against the register before flashing (recorded); only released baselines can be targeted by a released SU document; files only from the restricted Egnyte repository; suppliers do not deliver binaries for re-flash.	Baselines, verification entries	
7.1.3.2 Protection of the update process	Restricted repository access; SHA-256 cross-check; physical protection of vehicle update access points (covers, security bolts, tamper-evident seals); tool access control and audit trail.	5.4, 5.8	ISO 27001 evidence [TO BE CONFIRMED]
7.1.3.3 Verification and validation	V&V sign-off required before an SU document can be released.	SU report section 3	V&V procedure document reference [TO BE CONFIRMED]
7.1.4.1 / 7.1.4.2 OTA	Not applicable — no over-the-air updates (see scope statement).	Cover page, 2.2	

7 Open points

Items to be confirmed by eVersum before the handbook is released:

#	Topic	Question
1	RXSWIN on the vehicle	Are RXSWINs stored on the vehicle and readable via OBD? If not, the software versions must be declared to the Approval Authority and the declaration updated on every change (§7.2.1.2.2).
2	RXSWIN convention	Numbering format; one RXSWIN per regulation or per system; RXSWIN of the VCU (currently provisional VCUSWIN001).
3	Existing records	Export of the live Helix baselines or cut-off date; the checksums in the overview diagram are truncated placeholders.
4	System schemes baseline	Where it is held and the format of its identifier.
5	Type approval parameters	Which vehicle / system parameters must be recorded with the configuration (§7.1.2.2).
6	ERP	Identifier used by the ERP to link to the SUMS (currently the work order number); ERP read access to the Last Known Configuration is available.
7	Egnyte	Folder structure and file naming of the readme.
8	Users and authentication	Persons and roles; local accounts or AD / SSO.
9	Hosting, ISO 27001, backup	Production server, ISMS coverage, backup and restore policy.
10	Document structure	Alignment with the existing SUMS handbook / Dokumentenmanagement-Prozess chapter numbering; document owner and approvers.
11	Vehicle types	Vehicle types in scope of the first type approval; TÜV assessment date.
12	User notification	E-mail sent by the tool or only recorded.